Kurt Denver Chavez

kurtdenverchavez2005@gmail.com | linkedin.com/in/ kurt-chavez | github.com/kdenver14

Education

STI Ortigas Cainta — Cainta, Rizal
Bachelor of Science in Computer Science — *2027*

- Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Computer Architecture, Programming Methods, Web Development, Software Engineering

Experience

LOOP Marketers — Remote
Software Developer (Automation Project) — *Jan 2026 – Feb 2026*

- Built a Telegram automation script using the Telegram MTProto API and Telethon that listens for specific keywords in group chats and automatically responds in real time

react-hook-form — Remote
Open Source Contributor — *Dec 2025 – Jan 2026*

- Proposed a new useFormDebug hook to improve developer visibility into React Hook Form state during development
- Exposed complete formState including isDirty, isValid, errors, dirtyFields, and touchedFields
- Provided access to current form values, registered fields, and field arrays
- Implemented render count tracking to assist with performance debugging

Freelance — Antipolo City, Philippines
Freelance Developer – Build & Dependency Fixes — *Jan 2024 – Mar 2024*

- Fixed dependency conflicts in a TypeScript project and resolved complex build and tooling issues
- Troubleshoot Xcode configuration and environment problems to restore development workflows

Projects

Uptime Monitor | *TypeScript, Express 5, Prisma, PostgreSQL, Redis, BullMQ, Docker* — *Ongoing*

- Built a website monitoring backend with a versioned REST API for creating and managing monitored URLs
- Wrote a website checker that logs status code and response time, treating 4xx/5xx and timeouts as down
- Designed the architecture for scheduled checks on a Redis/BullMQ worker queue with email and Discord alerts
- Split the code into routes, controllers, services, and repositories to separate HTTP, logic, and data
- Validated requests with Zod, allowing only HTTP/HTTPS URLs and a minimum 30-second check interval
- Modeled data with Prisma ORM on PostgreSQL, running in Docker Compose with versioned migrations

Emotion & Sentiment Analysis for HR | *Next.js, TypeScript, PyTorch, Prisma, Modal* — *Aug 2025 – Oct 2026*

- Built a full-stack HR platform that analyzes interview and meeting videos for 7 emotions and 3 sentiments
- Trained a multimodal PyTorch model on MELD that fuses BERT text, R3D-18 video, and CNN audio features
- Served inference on Modal GPUs with Whisper and pyannote, using presigned R2 uploads and signed callbacks
- Added NextAuth.js login with emailed one-time codes, email verification, role-based access, and audit logs
- Built HR dashboards in Recharts for emotion trends and speaker timelines, with CSV and PDF exports

Int Wrapper for C++ | *C++, Memory Management, OOP* — *Mar 2024 – Dec 2025*

- Developed a robust custom Int class wrapper enhancing standard integers with OOP features and safety mechanisms
- Implemented full operator overloading for arithmetic and comparisons to match native integer behavior
- Engineered a ptr_deleter utility to safely handle recursive deletion of multiple pointers, preventing memory leaks

Technical Skills

Languages: TypeScript, JavaScript, Python, C++, C#, SQL
Frontend: React, Next.js, Tailwind CSS, Zustand, Vite
Backend: Node.js, Express, ASP.NET Core, NextAuth.js, REST APIs
Data: PostgreSQL, MySQL, SQLite, Prisma, Drizzle
ML: Keras, MediaPipe, OpenCV, NumPy
Tools: Git, GitHub, Docker, Vercel